

Mediation Analysis: Foundations and Application

Understanding Mechanisms and Indirect Effects

March 10, 2026

Introduction to GLM

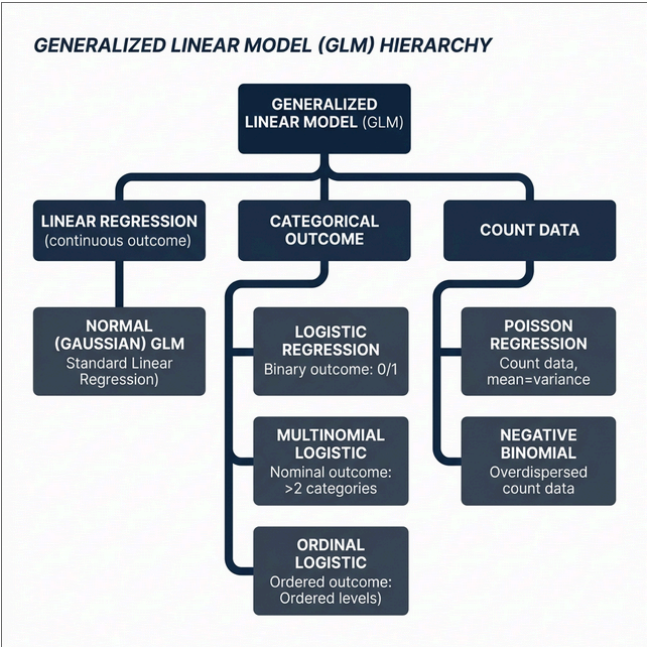
When we run statistical models in this course, most belong to the **Generalized Linear Model (GLM)** family. A **GLM** is a flexible statistical framework used to model relationships between variables under different types of outcomes.

GLM: Special Cases

Different regression models are **special cases of GLMs**, including:

- **Linear regression** (continuous outcomes)
- **Logistic regression** (binary outcomes)
- **Poisson regression** (count outcomes)

Goal: Explain the pathway connecting a predictor to an outcome.



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Mediation & the GLM

- Because these models share the same underlying framework, analytical tools such as **mediation analysis** can be applied within the GLM structure.
- **Mediation is not a separate type of model.**
- Instead, it is a way of using GLMs to understand **how a predictor influences an outcome through another variable.**

What is Mediation?

Traditional regression focuses on the **direct relationship**:
 $X \rightarrow Y$

Mediation asks a different question: > **What is the mechanism that links X to Y ?**

Structural Equation Foundations

- Mediation is the core of **Structural Equation Modeling (SEM)**.
- It allows us to build complex “**Roadmaps**” of behavior.
- **Modern Analysis:** We now use “Conditional Process Analysis” (Moderated Mediation).

Variable Roles in Mediation

The **Mediator** (M) is unique:

- It acts as a **Dependent Variable** because it is predicted by X .
- It acts as an **Independent Variable** because it predicts Y .

Summary

- X = Antecedent, Independent
- M = Intermediary (Mechanism)
- Y = Consequent, Dependent, Outcome

The Role of the Mediator (M)

- Instead of only estimating the direct effect, mediation examines the **process** connecting the variables.
- This process introduces an **intervening variable**, called the **mediator** (M).

The relationship becomes: $X \rightarrow M \rightarrow Y$

Direct vs. Indirect Effects

- **Direct Effect:** the effect of X on Y .
- **Indirect Effect:** the effect of X on Y through the mediator M .

Analogy: A mediator resolves a dispute by sitting between two parties. In statistics, M sits between the predictor (X) and the outcome (Y).

Introduction to Mediation

Mediation analysis is a statistical method used to evaluate how a causal **antecedent** variable (X) transmits its effect to a **consequent** variable (Y).

Rather than only asking whether X affects Y , mediation examines **the pathway through which the effect occurs**.

Research Question

What is the mechanism—emotional, cognitive, biological, or otherwise—by which X influences Y ?

Mediation as Causal Explanation

- Mediation provides a **causal explanation** of how X affects Y .
- Because of this, mediation models rely on **additional assumptions**.

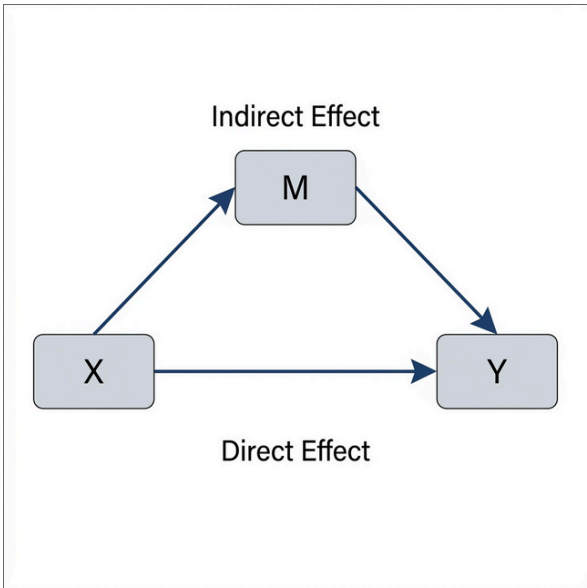
Causal Mediation Assumptions

- The relationships in the system are **causal**, not just correlational.
- The mediator (M) occurs **causally between** the predictor (X) and the outcome (Y).
- **The Causal Sequence:** $X \rightarrow M \rightarrow Y$.
- X influences M .
- M influences Y .

Important Considerations

- Important:*
When using **cross-sectional data**, establishing causal order can be difficult. Always consider whether the proposed mediator **logically occurs between X and Y** .
- Examples

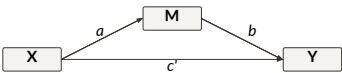
Mediation (Conceptual Model)



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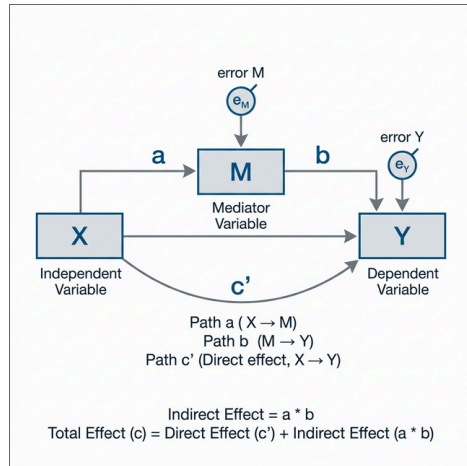
Mathematical Foundations

We treat mediation as two simultaneous regression equations:



1. M as the outcome ($M = i_M + aX + e_M$).
2. Y as the outcome ($Y = i_Y + c'X + bM + e_Y$).

Statistical Model Detail



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- **Path a**: How much M changes for a 1-unit change in X .
- **Path b**: The unique effect of M on Y (holding X constant).
- **Indirect Effect** ($a \times b$): The “Amount of Mediation.”

Summary of Key Terms

1. **Path a**: $X \rightarrow M$
2. **Path b**: $M \rightarrow Y$
3. **Indirect Effect**: $a \times b$
4. **Bootstrapping**: Resampling for significance.

The Statistical Model

Mediation is estimated using **two regression equations**.

Mediator model

$$M = i_M + aX + e_M$$

Outcome model

$$Y = i_Y + c'X + bM + e_Y$$

Key parameters

- **a**: Effect of X on the mediator (M)
- **b**: Effect of M on the outcome (Y), controlling for X
- **c'**: Direct effect of X on Y , controlling for M

The Statistical Model

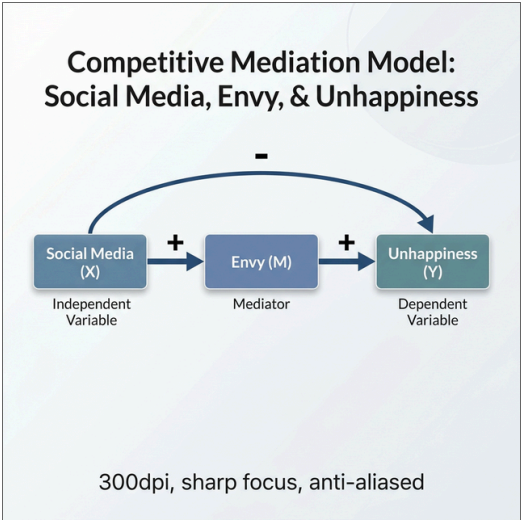
Goal

- Estimate the regression coefficients
- Combine them to evaluate the **indirect effect** ($a \times b$)

Important

Because we are predicting **two dependent variables**, mediation produces **two regression models**: - one predicting M - one predicting Y

Identify the Paths



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ab: Indirect path (Envy hurts happiness). **c'**: Direct path (Connecting with friends). This is a **Competitive Mediation** model.

Example: Education & Income

Suppose we want to understand **how education affects income**.

- X = Years of education
- M = Job skill level
- Y = Annual income

The Indirect Effect of X on Y

Definition	Details
Path a	How much two cases that differ by one unit on X are estimated to differ on M .
Path b	How much two cases that differ by one unit on M , but are equal on X , are estimated to differ on Y .
Indirect Effect (ab)	The product of a and b . It represents how much Y changes because X affects M .

Note: Interpretation depends on the **signs of a and b** .

Education Example: Path a

Path a (Education $\rightarrow$ Skills)
If two people differ by **one additional year of education**, path a tells us **how much higher their job skill level is expected to be**.

Education Example: Path a

Path a : Education $\rightarrow$ Job Skills

Suppose we estimate the regression:

$$M = 2 + 0.5X$$

where

- X = years of education
- M = job skill level

Interpretation

If **two people differ by one additional year of education**, the person with more education is expected to have **0.5 higher job skill units**.

Example:

Person	Education (X)	Predicted Skills (M)
A	12 years	8.0
B	13 years	8.5

Because the two people differ by **1 year of education**, their predicted skills differ by **0.5 units**.

Therefore:

Path $a = 0.5$.

Education Example: Path b

Path b (Skills $\rightarrow$ Income)

If two people differ by **one unit in job skill**, but have the **same level of education**, path b tells us **how much their income is expected to differ**.

Education Example: Path b

Path b : Skills $\rightarrow$ Income

Suppose we estimate the regression:

$$Y = 20,000 + 2,000M + 500X$$

where

- M = job skill level
- X = years of education
- Y = annual income

Interpretation

If **two people differ by one unit in job skill**, but have the **same level of education**, the person with higher skills is expected to earn **\$2,000 more per year**.

Example:

Person	Skills (M)	Education (X)	Predicted Income (Y)
A	8	12	\$42,000
B	9	12	\$44,000

Because the two people differ by **1 unit of skill**, their predicted income differs by **\$2,000**.

Therefore:

Path $b = 2000$.

Education Example: Indirect Effect

Indirect Effect (ab)

This tells us **how much income changes because education increases job skills**.

Education Example: Indirect Effect

Indirect Effect (ab)

The indirect effect tells us **how much income changes because education increases job skills**.

From the previous slides:

- **Path a :** One additional year of education increases job skills by **0.5 units**
- **Path b :** One additional unit of job skill increases income by **\$2,000**

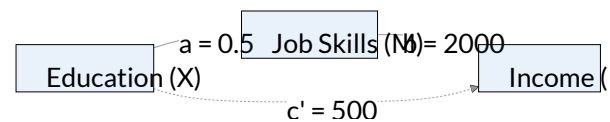
Calculate the indirect effect

$$ab = 0.5 \times 2000 = 1000$$

Interpretation

If **two people differ by one year of education**, the person with more education is expected to earn **\$1,000 more per year** through increased job skills.

Education Example: Mediation Results



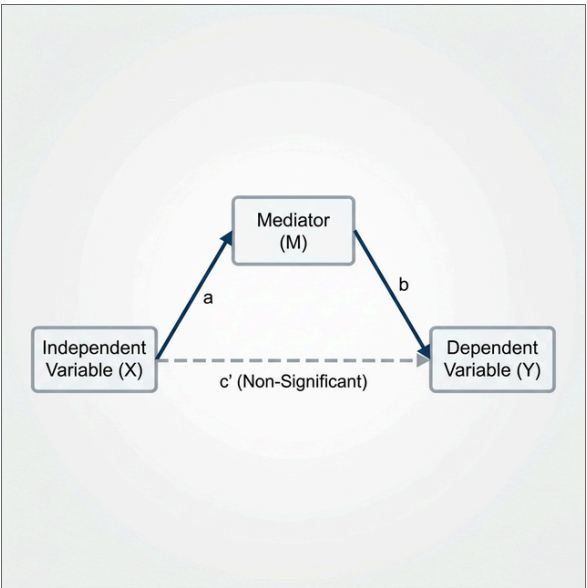
Interpretation

- **Path a :** Each additional year of education increases job skills by **0.5 units**.
- **Path b :** Each additional unit of job skill increases income by **\$2,000**.
- **Indirect effect (ab):** One additional year of education increases income by **\$1,000 through job skills**.
- **Direct effect (c'):** Education also increases income by **\$500 independent of job skills**.

The Total Effect of X on Y

Concept	Equation / Description
Total Effect	$c = c' + ab \rightarrow ab = c - c'$
Interpretation	Quantifies how much two cases that differ by one unit on X are estimated to differ on Y .
Path c	The change in Y for every 1-unit increase in X .
Difference	The difference between total and direct effect is that the direct effect controls for M .

Types of Mediation: Complete Mediation

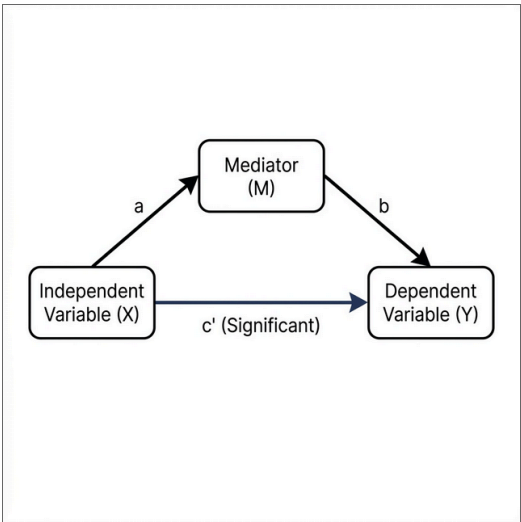


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Complete Mediation Defined

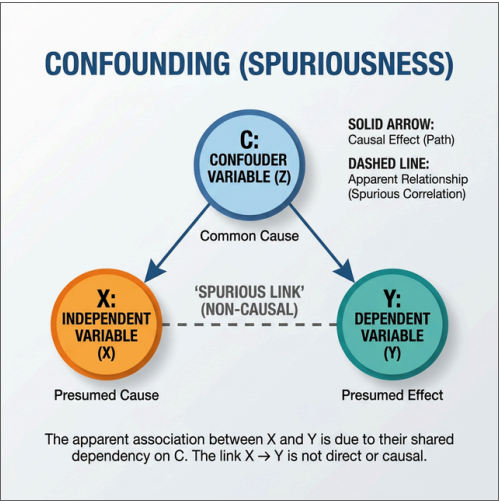
The relationship between X and Y is **totally explained** by the mediator. Without the mediator, the effect disappears.

Partial Mediation



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Partial Mediation: X affects Y both directly AND through the mediator. The mechanism explains *some* of the relationship, but not all of it.

Confounding vs. Mediating



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- **Confounder:** Creates a fake relationship between *X* and *Y*.
- **Mediator:** Explains a **real** relationship by showing the middle step.

Suppression Case Study: ADI

Variable	Without ADI	With ADI (Suppressor)
Growing Investment	0.782 (<i>ns</i>)	1.987 ($p < .05$)

Explanation: “Growing Investment” seemed unimportant until we accounted for neighborhood deprivation (ADI). ADI was hiding the real effect.

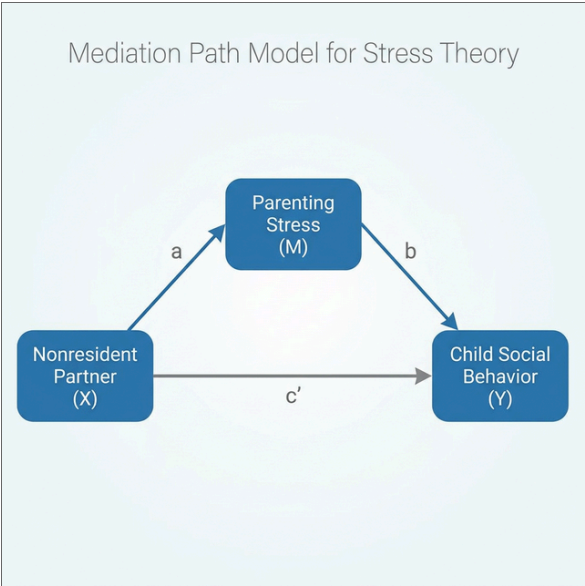
Suppression v Mediating Effects

- Suppression is “Mediation’s weird cousin.”
- Normally, adding a third variable makes the effect **smaller**, that is the relationship between *X* and *Y* is weaker.
- In **Suppression**, adding the third variable makes the effect **larger** and **more significant**.
- It works by “clearing the air” of noise (error variance).

Summary

1. **Full Mediation:** Direct path c' is zero.
2. **Partial Mediation:** Direct path c' is smaller but not zero.
3. **Suppression:** Effect of *X* on *Y* gets bigger when *M* is added to the model.
4. **Total Effect (c):** The sum of direct + indirect effects.

The "Stress Theory" Mechanism



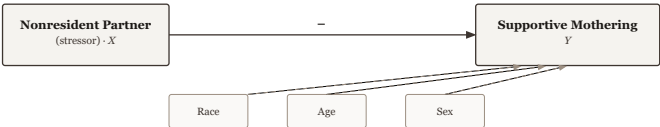
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Example: Parenting Stress & Child Behavior

Research Context

- **Observation:** Parenting by a mother with no partner is associated with a preschooler’s lower school readiness (social behavior).
- **Hypothesis:** This relationship is explained by the effects on mothers of the stress of a nonresident parenting partner.
- **The Mechanism:** Stress leads to less supportive parenting, which in turn affects children’s social behavior.

Simple Linear Regression (in a diagram)



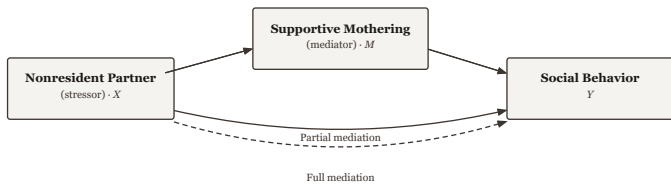
Dashed arrows = control variables (covariates)

Mediation Model Based on Stress Theory



Parenting by mother with no partner is associated with preschooler’s lower school readiness as measured by social behavior. The relationship is *explained* by the effects on mothers of the stress of a nonresident parenting partner. The stress is hypothesized to lead to less supportive parenting, which in turns affects children’s social behavior.

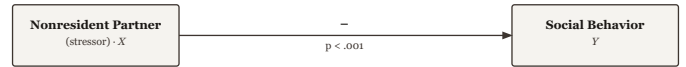
Mediation Model Based on Stress Theory



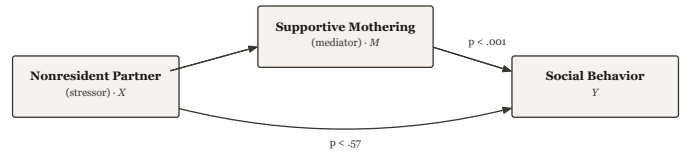
The dotted line indicates no significant direct effect (full mediation).

Full or Partial?

Step 1 – Direct effect (without mediator):



Step 2 – With mediator:

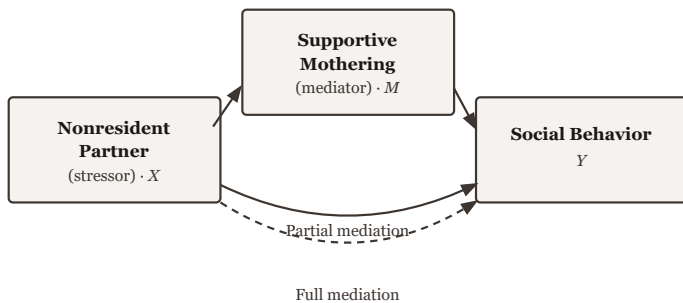


Identifying Mediation Through Regression Results

Regression equations have only one dependent variable. We can't test a mediation model in one analysis because two variables are dependent in the figure.

Which variables are dependent variables in the model (which are predicted by other variables)?

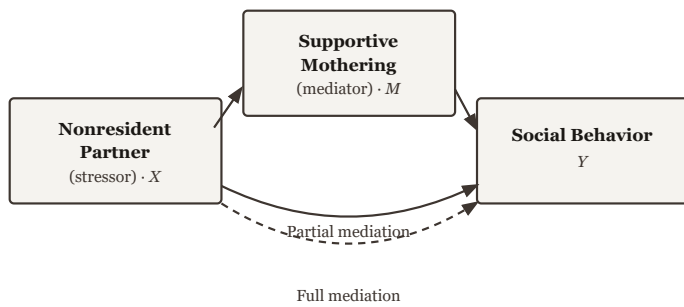
Which variable is both a predictor and a predicted variable in the figure?



How We Used to Test for Mediation in Regression?

Traditional Steps: Note that the distinction between full and partial is archaic.

1. Regress the DV on the IV and evaluate the IV's coefficient: (*Social Behavior on Nonresident Partner*)
2. Regress the mediator on the IV and evaluate coefficient: (*Supportive Mothering on Nonresident Partner*)
3. Regress the DV on both the IV and the mediator (*Social Behavior on Nonresident Partner and Supportive Mothering*)
4. Examine the magnitude and significance of the coefficients for both the IV and mediator
5. Test the significance of the two-part indirect path using the Sobel test



Practice: Full or Partial Mediation?

- **Step 1:** Before Mediator: Path c = Significant ($p < .001$).
- **Step 2:** After Mediator:
 - Path a = Significant.
 - Path b = Significant.
 - Path c' = Non-Significant ($p = .57$).

Answer: Full Mediation. The mediator “soaked up” all of X’s influence.

The Baron & Kenny 4-Step Method

1. **Step 1:** Establish that X and Y are related (Total Effect).
2. **Step 2:** Establish that X and M are related (Path a).
3. **Step 3:** Show that M predicts Y while controlling for X (Path b).
4. **Step 4:** Check if the $X \rightarrow Y$ relationship (c') disappeared.

Modern View on Step 1 (The Debate)

- **Old View (Baron & Kenny):** You CANNOT have mediation if the total effect (c) is not significant.
- **New View (Hayes):** You CAN have mediation without a significant total effect. The **Indirect Effect** is what fundamentally matters.

Process Macro vs. Manual Regression

- **Manual:** Run 2 regressions and multiply $a \times b$. Hard to test significance.
- **PROCESS** (Andrew Hayes): Automates the math and provides **Bootstrapping**, which journals now require.

Why Bootstrapping?

- Indirect effects (ab) are usually not “normally distributed.”
- Standard p-values assume a bell curve.
- **Bootstrapping** resamples data a large number of times to create a custom distribution and a **95% Confidence Interval**. There is NO p-value on the indirect effect.

Preparing R for Process

1. Download **process.R** from the Hayes website.
2. Run `source("process.R")` in your terminal.
3. Syntax: `process(subset_data, y="Y", x="X", m="M", model=4)`
4. **Model 4** = Simple Mediation.

Installing the PROCESS Macro in SPSS

The **PROCESS Macro** (developed by Andrew F. Hayes) allows SPSS to estimate **mediation, moderation, and conditional process models**.

Step 1: Download PROCESS

Go to the official website:

<https://www.processmacro.org/download.html>

Download the file **process.sps**.

Step 2: Open SPSS

1. Open **IBM SPSS Statistics**
2. Go to **File** → **Open** → **Syntax**
3. Open the downloaded file **process.sps**

Step 3: Install the Macro

1. With the syntax file open, click **Run** → **All**
2. SPSS will load the PROCESS macro into memory.

Step 4: Verify Installation

You should now see PROCESS in the menu:

Analyze → **Regression** → **PROCESS v4.x**

Important

PROCESS must be **loaded each time SPSS is restarted** unless you add it to your startup syntax.

Reference: Hayes, A. F. (2022). *Introduction to Mediation, Moderation, and Conditional Process Analysis*.

Numeric Problem: Study & Stress (Scenario)

Does stress mediate the association between study time and test score?

The Path: You study for 1 hour (X).

The Gain: Studying adds knowledge which increases score (+5 points, c').

The Cost: But studying also increases your stress (M).

Numeric Problem: The Mechanism

- X (Study) $\rightarrow$ M (Stress) $\rightarrow$ Y (Score)
- **Question:** Is the knowledge gained by studying (+5) cancelled out by the stress?
- Let's calculate the **Indirect Effect**.

Calculation: The Indirect Effect

- **Path a** = 2.0 (1 hr study = +2 units of stress)

- **Path b** = -3.0 (1 unit stress = -3 points on test)

Indirect Effect ($a \times b$): $2.0 \times -3.0 = -6.0$.

The stress created by studying hurts your test performance by 6 points.

Calculation: The Total Effect

The **Total Effect** (c) is the sum of the Direct and Indirect paths:

$$Total = Direct + Indirect$$

$$Total = 5.0 + (-6.0) = -1.0$$

Conclusion: Studying more actually **decreases** your score by 1 point because the stress is worse than the learning.

Question: **What do you notice about the story you told?

Visualizing the Tug-of-War

Effect	Path	Value	Result
Direct	c'	+5	Studying helps
Indirect	ab	-6	Stress hurts

Total c -1 Overall loss
Takeaway: X has competing effects on Y .

Case Study 1: The Sugar Shortage

The participants in this study (43 male and 80 female) read one of two newspaper articles describing an economic crisis that may affect the price and supply of sugar in Israel. Approximately half of the participants ($n = 58$) were given an article they were told would be appearing on the front page of a major Israeli newspaper (i.e., the “front-page” condition). The remaining participants ($n = 65$) were given the same article but were told it would appear in the middle of an economic supplement of this newspaper (i.e., the interior page condition).

PMI Study: Measures & Hypothesis

- The participants were asked about their reactions to the story including their intention to buy sugar (**REACTION**).
- They were also asked questions about how much they believed others would buy sugar in the community (*presumed media influence* measured by **PMI**).
- **Research Hypothesis:** Individuals who read the article on the front page would be more likely to think others would promptly buy sugar which in turn would lead them to buy sugar.

PMI: The Research Question

Does **Presumed Media Influence (PMI)** explain the relationship between article placement (Front vs. Back) and consumer reaction?

PMI Descriptive Statistics

Article Placement (X)	Mean PMI (M)	Mean Intent (Y)
Interior (0)	4.14	3.12
Front Page (1)	5.03	3.91

PMI Step 1: Article Placement on PMI

- **Predictor:** Condition (X).
- **Outcome:** PMI (M).

PMI Step 2: Predicting Reaction (Y)

- **Predictors:** Article Placement (X) and PMI (M).

OUTCOME VARIABLE: pmi

1	Model Summary:						
2	R	R-sq	MSE	F	df1	df2	p
3	.1808	.0327	1.7026	4.0878	1.0000	121.0000	.0454
4							
5	Model:						
6		coeff	se	t	p	LLCI	ULCI
7	constant	5.3769	.1618	33.2222	.0000	5.0565	5.6973
8	cond	.4765	.2357	2.0218	.0454	.0099	.9431

- **Finding:** Article placement has a **significant positive effect** on PMI.
- **Regression Weight:** $\beta = 0.4765, p = 0.0454$.
 - Front page placement significantly increases perceived media influence.
- **Model Fit:** $R^2 = 0.0327$.
- **Test Statistic:** $F(1, 121) = 4.09, p = .045$.
Interpretation: Front-page placement significantly increases PMI by .477 units.

PMI: Raw PROCESS Output (Model Y)

OUTCOME VARIABLE: reaction

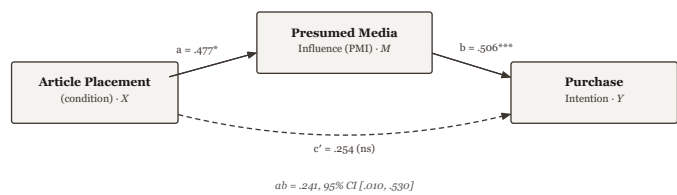
1	Model Summary:						
2	R	R-sq	MSE	F	df1	df2	p
3	.4538	.2059	1.9404	15.5571	2.0000	120.0000	.0000
4							
5	Model:						
6		coeff	se	t	p	LLCI	ULCI
7	constant	.5269	.5497	.9585	.3397	-.5615	1.6152
8	cond (c')	.2544	.2558	.9943	.3221	-.2522	.7609
9	pmi (b)	.5064	.0970	5.2185	.0000	.3143	.6986

- Model Fit: $R^2 = 0.2059$ ($F(2, 120) = 15.56, p < .001$).
- PMI Results: Significantly predicted reaction ($\beta = 0.5064, p < .001$).

PMI: Preliminary Findings

- Both M (Presumed Media Influence) and Y (Buying Intentions) scores increase when the article is on the front page.
- This suggests a potential mediation effect to investigate.

PMI: Statistical Mediation Model



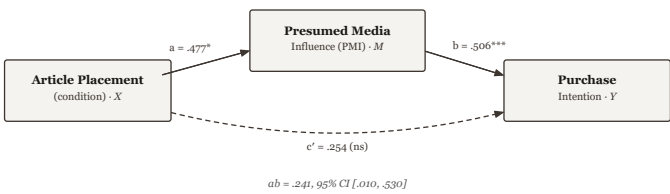
Model M – Outcome: PMI

	coeff	p
Constant	5.377	.000

Condition (a) **0.477** **.045**

$R^2 = .033, F(1, 121) = 4.09, p = .045$

PMI: Statistical Model (cont)



Model Y – Outcome: Reaction

	coeff	p
Constant	0.527	.340
Condition (c')	0.254	.322

PMI (b) **0.506** **<.001**

$R^2 = .206, F(2, 120) = 15.56, p < .001$

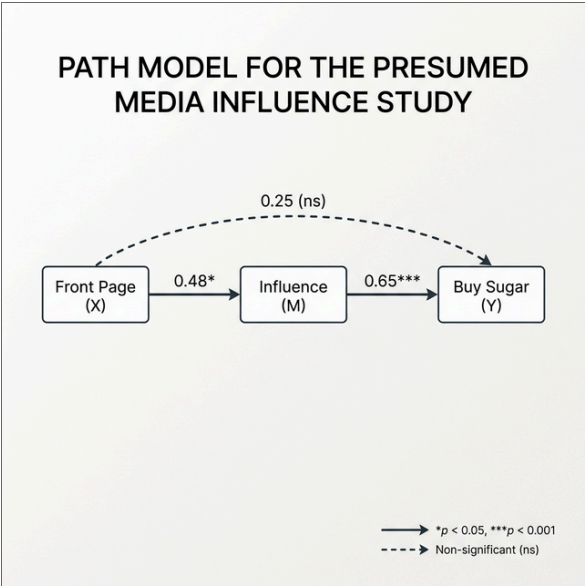
Dashed path = non-significant direct effect. Indirect effect: $ab = .241, 95\% \text{ CI } [.010, .530]$ – significant (excludes zero).

Step 3: Direct vs. Indirect Effects (PMI)

Effect Type	Coefficient	95% Confidence Interval	Result
Direct Effect (c')	0.2544	[-0.2522, 0.7609]	Non-Significant
Indirect Effect (ab)	0.2413	[0.0095, 0.5304]	Significant

- **Placement Result:** Did **not** significantly predict reaction ($\beta = 0.2544, p = .322$).
- **Key Takeaway:** Article placement alone does not directly impact reaction; it requires the belief about others (mediator).
- **Conclusion:** Since the indirect effect interval does not contain zero, mediation occurred.
- We have evidence for the mechanism.

PMI Summary Diagram



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PMI: The Final Verdict

- **Result: Full Mediation.**
- The location only works through the belief about the crowd.
- X influence is entirely absorbed by M .
- **Mechanism Identified:** PMI is the key explanatory factor.
- **Practical Application:** Enhancing perceived influence may be more impactful than simple placement in a newspaper.
-

Narrative Summary: PMI Results

- **Path a :** Placement significantly predicted PMI ($a = 0.477, p = .045$).
- **Path b :** PMI significantly predicted purchase intention ($b = 0.506, p < .001$).
- **Path c' :** Direct effect was non-significant ($c' = 0.254, p = .322$).

Study 1 Final Conclusion

- The front page story worked by triggering a specific psychological state.
- **Mechanism:** Triggered a belief that others would hoard sugar.
- This perceived threat drives the actual consumer reaction.

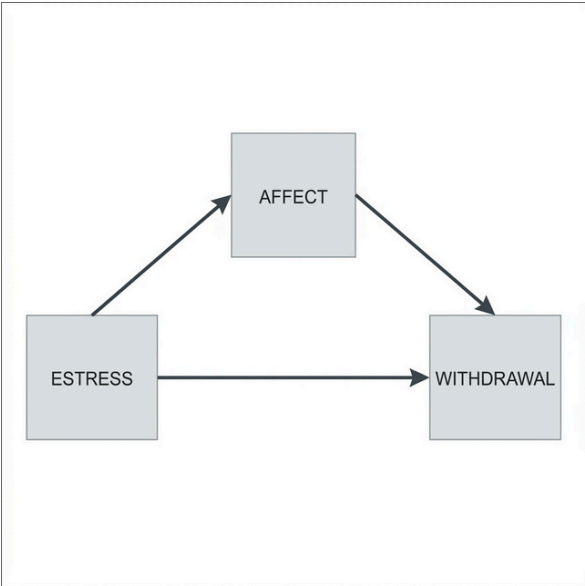
Case Study 2: Economic Stress (ESTRESS)

Participants in this study were $N = 262$ entrepreneurs who were members of Business Networking International, a networking group for small-business owners, who responded to an online survey about recent performance of their business as well as their emotional and cognitive reactions to the economic climate.

ESTRESS Study: Measures & Hypothesis

- Participants were asked about how their business was doing (the measure of economic stress, **ESTRESS**) along with their feelings related to their business (**AFFECT**) and intentions to withdrawal from being a business owner (higher scores == more withdrawal intentions).
- **Hypothesis:** The effect of economic stress (X) on disengagement from entrepreneurial activities (Y) operates through depressed affect (M).

Detailed Model: Economic Stress



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ESTRESS: Raw PROCESS Output (Model M)

OUTCOME VARIABLE: affect

1 Model Summary:							
2	R	R-sq	MSE	F	df1	df2	p
3	.3401	.1156	.4650	33.9988	1.0000	260.0000	.0000
4							
5	Model:						
6		coeff	se	t	p	LLCI	ULCI
7	constant	.7994	.1433	5.5777	.0000	.5172	1.0816
8	estress (a)	.1729	.0296	5.8308	.0000	.1145	.2313

ESTRESS: Raw PROCESS Output (Model Y)

OUTCOME VARIABLE: withdraw

1 Model Summary:							
2	R	R-sq	MSE	F	df1	df2	p
3	.4247	.1804	1.2841	28.4946	2.0000	259.0000	.0000
4							
5	Model:						
6		coeff	se	t	p	LLCI	ULCI
7	constant	1.4471	.2520	5.7420	.0000	.9508	1.9433
8	estress (c')	-.0768	.0524	-1.4667	.1437	-.1800	.0263
9	affect (b)	.7691	.1031	7.4627	.0000	.5662	.9721

Results - Outcome Variable: Affect (Path a)

Model Summary	R	R-sq	F	p
	0.3401	0.1156	33.99	.0000
Predictor	coeff	se	t	p
Constant	0.7994	0.14	5.57	.0000
ESTRESS (a)	0.1729	0.02	5.83	.0000

Interpretation: Effect on Affect

- **Finding:** Economic stress has a significant positive effect on depressed affect.
- **Coefficient:** $\beta = 0.1729, p < .001$.

Interpretation: Economic Stress Scale

- **Scale:** Every 1-unit increase in economic stress increases affect by 0.1729 units.
- **Meaning:** As economic stress increases, depressed affect also increases.

Results: Predicting Withdrawal (Path b & c')

Model Summary

R	R-sq	MSE	F	p
0.4247	0.1804	1.2841	28.4946	.0000

Model Coefficients

Predictor	coeff	se	t	p
Constant	1.4471	0.25	5.74	.0000
ESTRESS (c')	-0.0768	0.05	-1.47	.1437
AFFECT (b)	0.7691	0.10	7.46	.0000

Recall: c' is the direct effect.

Interpretation: Predicting Withdrawal

- Model Fit: The model explained 18.04% of the variance in withdrawal ($R^2 = 0.1804$).
- Significant Predictor: Affect significantly predicts withdrawal ($\beta = 0.7691, p < .001$).

Interpretation: Economic Stress (ESTRESS) Path

- Non-Significant Predictor: Economic Stress was not a significant predictor after accounting for affect ($\beta = -0.0768, p = 0.1437$).
- Insight: Higher affect leads to greater withdrawal, but economic stress does not directly predict withdrawal.

Decomposing the Effects (Study 2)

Effect Type	Coefficient	Interpretation
Total Effect (c)	0.0561	$p = 0.3015$ (Non-significant)
Direct Effect (c')	-0.0768	$p = 0.1437$ (Non-significant)
Indirect Effect (ab)	0.1330	Significant (Mediation)

Note: Total effect (c) = $c' + ab \rightarrow -0.0768 + 0.1330 = 0.0561$. Economic stress alone does not predict withdrawal; the effect is **fully mediated** by affect.

The Indirect Effect (ab): Statistical Analysis

Predictor	Effect	BootSE	BootLLCI	BootULCI
Affect (ab)	0.1330	0.0335	0.0694	0.2015

Statistical Significance (ab)

- **Significant:** The 95% CI [0.0694, 0.2015] **does not contain zero**.
- **Meaning:** Every 1-unit increase in economic stress leads to a 0.133 increase in withdrawal intentions *through* depressed affect.
- The effect is moderate and reliable.

Why the Total Effect “Disappeared”

- **Direct Path (c'):** Negative (-0.08)
- **Indirect Path (ab):** Positive (+0.13)
- **Total Effect (c):** +0.05 ($p = .30$).

ESTRESS: The Balancing Act

- **Explanation:** Stress makes you want to quit *because* of depression, but resilience might keep you there.
- These opposing forces cancel each other out in the total effect.

Conclusion: Economic Stress Example

- **Mechanism:** Economic stress increases depressed affect, which in turn increases withdrawal behavior.
- **Full Mediation:** The relationship between economic stress and withdrawal is **fully mediated** by affect.
- **Key Takeaway:** Stress only leads to withdrawal through its emotional toll (business affect).

Resources for Further Study

- **Hayes (2017):** *Introduction to Mediation, Moderation, and Conditional Process Analysis*.
- **Preacher & Hayes (2004):** The seminal paper on Bootstrapping.

Q&A and Final Review

- See Worksheet

Speaker notes